

MEDS Lab Module 4

Section 10: Counters & Timing Applications

Task 12: Modulo-10 Frequency Divider

July 23, 2026

1. EDA Playground Link

The complete design and testbench code for the Modulo-10 Frequency Divider can be found at the following shareable link:

- <https://edaplayground.com/x/kdAu>

2. Schematic Diagram

Below is the block diagram illustrating the modulo-10 wrapping counter and the T flip-flop output stage.

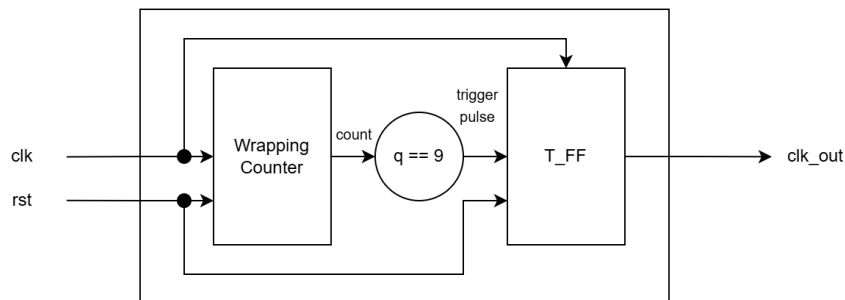


Figure 1: Block diagram of the modulo-10 frequency divider.

3. Terminal Output Screenshot

The following screenshot displays the simulation log, confirming that the self-checking testbench successfully counted exactly 10 input clock edges between the output edges.

```

Compiler version X-2025.06-SP1_Full164; Runtime version X-2025.06-SP1_Full164;  Jul 23 09:49 2026
PASS: Exactly 10 input clocks counted between output edges.
$finish called from file "testbench.sv", line 28.
$finish at simulation time                205
      V C S   S i m u l a t i o n   R e p o r t

```

Figure 2: Simulation log output from EDA Playground verifying the divide-by-10 ratio.

4. Waveform Screenshot

The waveform below visualizes the clock division, demonstrating the counter tracking from 0 to 9 and the output clock toggling accordingly.

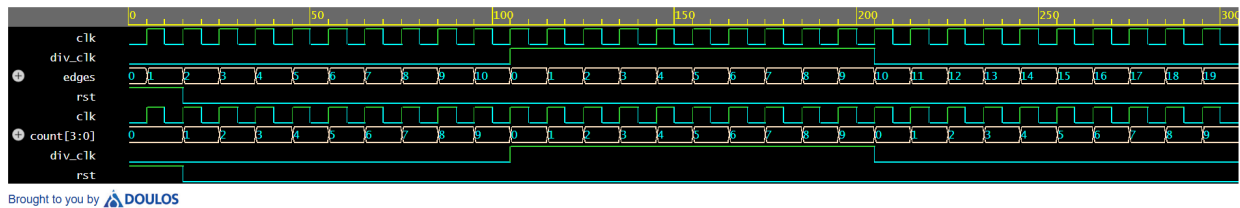


Figure 3: EPWave waveform confirming the modulo-10 wrapping and clock division.

5. Explanation

This task involved designing a modulo-10 frequency divider that reduces the input clock frequency by a factor of 10. The implementation uses a 4-bit counter to track the input clock cycles. Upon reaching a terminal count of 9, the counter is reset to 0 on the next active clock edge. Simultaneously, this terminal count condition toggles the `div_clk` output, effectively modeling the behavior of a T flip-flop. The provided self-checking testbench validates this logic by verifying that exactly 10 input clock edges occur for every transition of the divided output clock.